SATHISH G

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EXECUTIVE SUMMARY

AI Full-Stack Engineer with 2.5+ years of experience building production-grade conversational AI systems and scalable backend platforms. Specialized in RAG-based LLM pipelines, real-time streaming architectures, and cloud-native microservices deployed on Kubernetes (AWS EKS, Azure AKS). Experienced in delivering end-to-end AI products from frontend interfaces to backend orchestration serving 50K+ daily users.

SKILLS

Languages:	Python, JavaScript, TypeScript, SQL, Java
AI/LLM Systems:	LangChain, LangGraph, OpenAI API, RAG Pipelines, Prompt Engineering, Vector Search, NLP
Backend:	FastAPI, Flask, Spring Boot, REST APIs, WebSocket, Microservices Architecture
Frontend:	React, Angular, Electron
Cloud Infrastructure:	AWS (EKS, S3, Lambda), Azure (AKS, ADLS, Cognitive Services), Kubernetes, Docker, GitHub Actions
Databases:	PostgreSQL, MongoDB, Elasticsearch, Redis

AI ARCHITECTURE HIGHLIGHTS

- Designed Retrieval-Augmented Generation (RAG) pipelines with vector-based semantic search and contextual memory.
- Implemented real-time streaming LLM responses using WebSockets for latency-sensitive applications.
- Built modular AI microservices enabling independent scaling and experimentation.
- Optimized prompt templates and token usage for cost-efficient and reliable AI outputs.

EXPERIENCE

Software Engineer — Jul 2023 - Present
Medha Analytics — *Bangalore, Karnataka*

Call Center AI Copilot — Python, Node.js, LangChain, React, WebSocket

- Designed a real-time AI copilot for customer care agents processing dual-channel audio with sub-500ms response latency.
- Built a RAG-based conversational pipeline using LangChain with contextual retrieval for live agent assistance.
- Architected 6 microservices (Python + Node.js) with independent scaling and centralized observability.
- Developed audio processing pipeline with VAD and multi-channel separation for simultaneous transcription.

DataChat AI – Healthcare Analytics Platform — Python, LangChain, FastAPI, SQL

- Built an NLP-to-SQL conversational system converting natural language queries into optimized SQL statements.
- Implemented structured output validation and query safety checks to improve SQL reliability.
- Designed caching and query optimization mechanisms supporting 500+ daily analytical queries.

Appointment Mitra – WhatsApp AI Assistant — Python, WhatsApp API, LangChain

- Developed a WhatsApp-based conversational AI assistant for automated appointment booking.
- Designed state-machine driven multi-step conversation workflows with context tracking.

- Integrated geolocation-based hospital matching, availability checks, and payment gateway.

Kube Board – Multi-Cloud Kubernetes Platform Flask, Python, Kubernetes

- Built a full-stack SaaS platform for centralized Kubernetes cluster management across AWS and Azure.
- Implemented OAuth2 authentication and RBAC with real-time log streaming using Elasticsearch.
- Developed scheduled resource optimization strategies to reduce infrastructure waste.

Enterprise CI/CD Pipeline GitHub Actions, AWS EKS, Docker

- Designed CI/CD workflows reducing deployment time from 45 minutes to 12 minutes.
- Implemented blue-green deployment strategy with automated health checks and rollback mechanisms.
- Integrated automated test gates (unit, integration, security scans) to improve production stability.

PROJECTS

Food Demand Forecasting System Python, Flask, XGBoost, LSTM

- Built predictive analytics models (Random Forest, XGBoost, LSTM) for meal demand forecasting.
- Developed REST APIs and dashboard for real-time visualization and inventory optimization.

EDUCATION

Bachelor of Computer Science 2019 - 2023
Government College of Engineering, Erode
CGPA: 8.18 / 10

CERTIFICATIONS

- System Design: Low-Level to High-Level Solutions – GeeksforGeeks
- Generative AI Applications Using Amazon Bedrock – AWS
- Python for Data Science – NPTEL (IIT)